Supplementary Material: Physiology-Guided Temporal Fusion for Cross-Subject EEG–fNIRS Brain–Computer Interfaces

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This supplementary document provides extended experimental analyses, physiological interpretations, and subject-level ablation breakdowns supporting the main manuscript:

1) **Section S.I:** Comparative neurovascular coupling dynamics and delay sensitivity analysis between cognitive (mental arithmetic) and sensorimotor (motor imagery) paradigms (Supplementary Fig. S1).
2) **Section S.II:** Subject-by-subject (29 folds) factorial prior retraining breakdown on motor imagery (Supplementary Table S1).
3) **Section S.III:** Mathematical formulation and streaming convergence dynamics of Causal Recursive Euclidean Alignment (CrEA).
4) **Section S.IV:** Computational reproducibility parameters and training environment specifications (Supplementary Table S2).
5) **Section S.V:** Independent external cohort cross-subject validation on the 15-participant Buccino et al. (2016) benchmark (Supplementary Table S3).

S.I. Paradigm-Specific Neurovascular Coupling Dynamics

A. Contrast Between Cognitive and Sensorimotor Paradigms

Cross-paradigm evaluation (Section V-F of the main manuscript) indicates that neurovascular coupling manifests distinct functional characteristics across task regimes: metabolic-dominant cognitive processing (Mental Arithmetic: 79.37%; Working Memory: 73.24%) benefits from structured hemodynamic delay priors, whereas sensorimotor tasks (Motor Imagery: 66.32%) primarily engage rapid transient electrophysiological desynchronization (μ/β ERD). To characterize this physiological transition and evaluate delay sensitivity across paradigms, we conducted two investigations on the 29-participant motor imagery dataset from Shin et al. (2017):

1) **Fixed-Weight Candidate Delay Sweep ($\tau \in [0.0, 8.0]$ s):** Evaluating the sensitivity of the trained cross-modal attention network to varying candidate delay parameters without retraining.
2) **2×2 Factorial Prior Retraining Ablation:** Full model retraining across 29 LOSO folds under four permutations of the Gaussian delay prior $\mathbf{P}$ and the discrete causal mask $\mathbf{M}$ ($P_{\text{on}}, M_{\text{on}}$; $P_{\text{on}}, M_{\text{off}}$; $P_{\text{off}}, M_{\text{on}}$; $P_{\text{off}}, M_{\text{off}}$).

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B. Empirical Findings and Biophysical Mechanism

As shown in Supplementary Fig. S1(a), mental arithmetic exhibits delay sensitivity: accuracy increases monotonically from 75.34% at $\tau = 0.0$ s to plateau across the physiological window ($\tau \in [3.5, 6.5]$ s), reaching 80.46% at $\tau = 8.0$ s ($\Delta = +5.12$ pp, $p < 0.001$). In contrast, motor imagery yields a flat trajectory: grand mean accuracy fluctuates between 68.91% ($\tau = 1.0$ s) and 69.60% ($\tau = 8.0$ s), an amplitude range of 0.69 pp, with overlapping standard error intervals across all candidate latencies.

The retrained factorial decomposition in Supplementary Fig. S1(b) indicates a paradigm-dependent disparity. For mental arithmetic, omitting both priors ($P_{\text{off}}, M_{\text{off}}$) reduces accuracy by 4.94 pp (79.37% $\rightarrow$ 74.43%, $p = 6.05 \times 10^{-5}$), with the Gaussian prior accounting for an average marginal contribution of +4.31 pp. In contrast, for motor imagery, retraining with both priors disabled yields 68.28% $\pm$ 13.72% vs. 69.20% $\pm$ 14.40% for the full model ($\Delta = -0.92$ pp, Wilcoxon $W = 106.5$, $p = 0.2128$, non-significant). Omitting causal masking alone ($P_{\text{on}}, M_{\text{off}}$: 68.10%, $p = 0.0978$) or the delay prior alone ($P_{\text{off}}, M_{\text{on}}$: 68.68%, $p = 0.1766$) likewise shows no statistically significant divergence.

Biophysically, this contrast aligns with established physiological distinctions in functional vascular dynamics reported in the neuroimaging literature:

1) **Prefrontal Cognitive Workload (MA):** Sustained 10-second mental arithmetic imposes metabolic demands on the prefrontal cortex, recruiting arteriolar dilation and venous drainage volume [1], [2]. This produces frontal [HbO] elevations (typically in the 0.2–0.6 μM range in continuous-wave fNIRS recordings [3]) that evolve over 4–7 s and persist throughout task engagement. Consequently, cross-modal attention relies on physiological delay priors to align frontal theta oscillations (θ ERS, 4–8 Hz) with delayed capillary hyperoxygenation, preventing acausal fitting on held-out participants.
2) **Sensorimotor Lateralization (MI):** Kinesthetic motor imagery elicits immediate contralateral sensorimotor rhythm desynchronization (μ and β ERD, 8–30 Hz) over primary motor cortex (C3/C4). However, imaginary finger tapping without physical contraction produces lower and more diffuse hemodynamic changes (typically

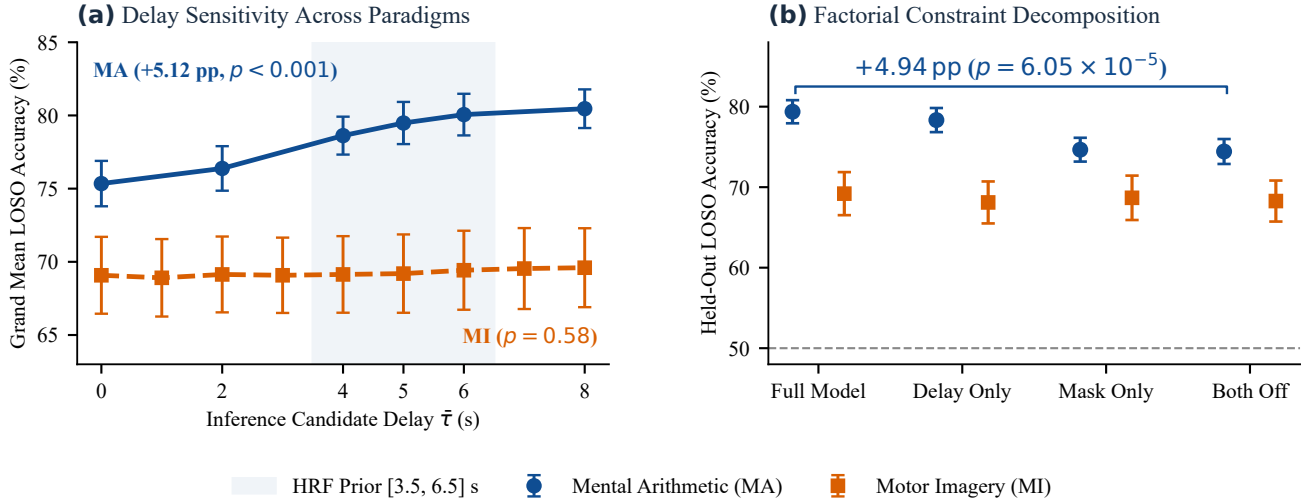


Fig. S1. Comparative neurovascular delay sensitivity and prior ablation contrast between cognitive (Mental Arithmetic, MA) and sensorimotor (Motor Imagery, MI) paradigms across 29 participants under participant-isolated LOSO cross-validation. (a) Candidate delay sensitivity scan across $\tau \in [0.0, 8.0]$ s without retraining. Error bars denote standard error of the mean (SE across $N = 29$ folds). For MA, accuracy rises monotonically from 75.34% at $\tau = 0.0$ s to plateau across the physiological hemodynamic response window ($\tau \in [3.5, 6.5]$ s) and reaches 80.46% at $\tau = 8.0$ s (+5.12 pp, $p < 0.001$), reflecting the sustained 10.0 s task block duration. For MI, the performance curve exhibits delay-invariant behavior within observed experimental variation (ranging between 68.91% and 69.60%, $\Delta \leq 0.69$ pp, p = n.s.). (b) Points and error bars show means and SEM across 29 folds; the dashed baseline marks 50% chance accuracy. Retrained 2×2 factorial decomposition of delay prior $\mathbf{P}$ and causal mask $\mathbf{M}$. On MA, dropping both priors causes a -4.94 pp performance reduction ($p = 6.05 \times 10^{-5}$), consistent with the regularizing role of neurovascular constraints. On MI, removing both priors decreases accuracy by 0.92 pp ($p = 0.2128$, non-significant), indicating that unconstrained attention performs similarly when hemodynamic coupling is weakly localized.

0.05–0.15 μM [3], [4]). Because optical signal-to-noise ratio is lower and neural decodability is dominated by instantaneous spectral ERD, the hemodynamic delay parameter τ provides limited regularization. Consequently, unconstrained cross-attention and direct spatial filtering models (such as FGANet) perform similarly.

S.II. Subject-by-Subject Factorial Prior Breakdown

Supplementary Table S1 lists the individual test accuracy achieved across each of the 29 held-out participant folds on motor imagery under all four factorial retraining conditions. This dedicated 29-fold factorial retraining experiment was conducted independently to isolate the individual contributions of the delay prior and causal mask (yielding 69.20% $\pm$ 14.40% for the full model), alongside the initial cross-paradigm benchmark reported in Table V of the main manuscript (66.32% $\pm$ 12.18%). In this factorial experiment, removing both temporal priors reduced MI accuracy by 0.92 pp ($p = 0.2128$), compared with a 4.94 pp reduction on mental arithmetic.

Between-subject standard deviations ($\sim$ 14.0 pp) substantially exceed the marginal differences observed across model conditions ($\leq$ 1.1 pp). High responders (e.g., Sub-19 and Sub-26: $> 90\%$) decode accurately regardless of prior conditioning, while BCI-illiterate participants (e.g., Sub-10: 45.00%, Sub-24: 50.00%, Sub-12: 51.67%) decode near chance level.

S.III. Causal Recursive Euclidean Alignment (CREA)

Standard Euclidean Alignment (EA) requires computing the arithmetic Euclidean sample mean covariance matrix across

TABLE S1
PARTICIPANT-BY-PARTICIPANT TEST ACCURACY (%) ACROSS 29 LOSO FOLDS ON MOTOR IMAGERY UNDER 2×2 FACTORIAL RETRAINING OF DELAY PRIOR $\mathbf{P}$ AND CAUSAL MASK $\mathbf{M}$.

Participant Fold	Full Model ($P_{\text{on}}, M_{\text{on}}$)	Mask Only ($P_{\text{off}}, M_{\text{on}}$)	Delay Only ($P_{\text{on}}, M_{\text{off}}$)	Both Off ($P_{\text{off}}, M_{\text{off}}$)	Δ vs. Full (Both Off $-$ Full)
Sub-01	80.00	80.00	81.67	80.00	+0.00
Sub-02	68.33	63.33	65.00	60.00	−8.33
Sub-03	73.33	75.00	73.33	71.67	−1.67
Sub-04	56.67	55.00	60.00	58.33	+1.67
Sub-05	56.67	55.00	56.67	56.67	+0.00
Sub-06	68.33	65.00	68.33	66.67	−1.67
Sub-07	55.00	55.00	53.33	51.67	−3.33
Sub-08	70.00	90.00	73.33	76.67	+6.67
Sub-09	81.67	83.33	81.67	78.33	−3.33
Sub-10	45.00	51.67	46.67	53.33	+8.33
Sub-11	76.67	71.67	73.33	76.67	+0.00
Sub-12	51.67	45.00	45.00	50.00	−1.67
Sub-13	65.00	66.67	63.33	58.33	−6.67
Sub-14	63.33	65.00	65.00	66.67	+3.33
Sub-15	66.67	61.67	65.00	61.67	−5.00
Sub-16	90.00	88.33	85.00	86.67	−3.33
Sub-17	86.67	88.33	83.33	85.00	−1.67
Sub-18	53.33	51.67	53.33	51.67	−1.67
Sub-19	98.33	91.67	91.67	90.00	−8.33
Sub-20	63.33	60.00	63.33	60.00	−3.33
Sub-21	71.67	71.67	70.00	76.67	+5.00
Sub-22	61.67	58.33	61.67	58.33	−3.33
Sub-23	68.33	68.33	68.33	70.00	+1.67
Sub-24	50.00	50.00	48.33	50.00	+0.00
Sub-25	90.00	93.33	93.33	91.67	+1.67
Sub-26	98.33	96.67	98.33	98.33	+0.00
Sub-27	80.00	73.33	71.67	73.33	−6.67
Sub-28	53.33	53.33	51.67	55.00	+1.67
Sub-29	63.33	63.33	63.33	66.67	+3.33
Mean	**69.20**	**68.68**	**68.10**	**68.28**	**−0.92**
SD	14.40	14.85	14.03	13.72	4.10
Macro F1	67.00	66.26	65.79	66.06	−0.94
Wilcoxon p	–	0.1766	0.0978	0.2128	–

Evaluated on the 29-participant Motor Imagery benchmark from Shin et al. (2017) [3] across 1,740 trials (60 trials per participant, 30 Left Hand vs. 30 Right Hand). All variants trained from scratch under identical hyperparameter schedules. Wilcoxon signed-rank p-values are computed against the Full Model ($N = 29$ participants).

the entirety of a testing session:

$$\overline{\mathbf{R}}_{\text{test}} = \frac{1}{K}\sum_{k=1}^{K}\mathbf{C}_k, \tag{S1}$$

where $\mathbf{C}_k = \frac{1}{T_e-1}\mathbf{X}_k\mathbf{X}_k^\top + \epsilon\mathbf{I} \in \mathbb{R}^{C\times C}$ is the regularized sample spatial covariance of trial k (with $\epsilon = 10^{-5}$), and $\mathbf{X}_k \in \mathbb{R}^{C\times T_e}$ is the temporal mean-centered EEG trial matrix ($\sum_{t=1}^{T_e} X_{k,c,t} = 0$ for each channel c). Although effective for offline validation, (S1) relies on transductive lookahead, precluding causal real-time operation at the bedside.

To eliminate session lookahead, Causal Recursive Euclidean Alignment (CrEA) tracks spatial covariance online via recursive empirical Bayes shrinkage:

$$\mathbf{R}_k = \frac{\kappa_0\mathbf{R}_{\text{src}} + \sum_{m=1}^{k}\mathbf{C}_m}{\kappa_0 + k}, \quad k = 1, 2, \ldots, K, \tag{S2}$$

where $\mathbf{R}_{\text{src}} = \frac{1}{N_{\text{tr}}}\sum_{s\in\mathcal{S}_{\text{tr}}}\overline{\mathbf{R}}_s$ is the reference covariance estimated offline exclusively from the $N_{\text{tr}} = 23$ training participants, and $\kappa_0 = 5.0$ represents the pseudo-observation strength of the source prior. In the diagnostic convergence protocol (Supplementary Fig. S2), $\mathbf{R}_{\text{src}}$ is scaled to match the trace of the target participant's first trial via $\widetilde{\mathbf{R}}_{\text{src}} = \mathbf{R}_{\text{src}} \cdot \frac{\text{tr}(\mathbf{C}_1)}{\text{tr}(\mathbf{R}_{\text{src}})}$ to harmonize the power scale. The comparison identity baseline is similarly trace-matched via $\widetilde{\mathbf{R}}_0^{\text{identity}} = \frac{\text{tr}(\mathbf{C}_1)}{C}\mathbf{I}$. In the streaming classification pipeline (71.21% accuracy across 29 folds), unscaled $\mathbf{R}_{\text{src}}$ is used directly (under which the 8th-trial relative covariance error reaches 36.47%).

The online whitening matrix is obtained via eigendecomposition $\mathbf{R}_k = \mathbf{V}_k\mathbf{\Lambda}_k\mathbf{V}_k^\top$:

$$\mathbf{W}_k = \mathbf{R}_k^{-1/2} = \mathbf{V}_k\,\text{diag}\left(\lambda_{k,c}^{-1/2}\right)\mathbf{V}_k^\top. \tag{S3}$$

The aligned EEG trial is transformed causally as $\widetilde{\mathbf{X}}_k = \mathbf{W}_k\mathbf{X}_k$.

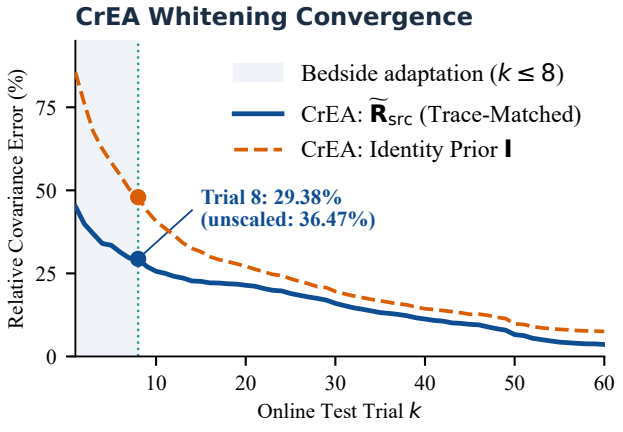


Fig. S2. Online Causal Recursive Euclidean Alignment (CrEA) covariance whitening convergence dynamics across 60 continuous trials ($k = 1, \ldots, 60$) averaged across all 29 held-out participants under the diagnostic protocol. The trace-matched empirical Bayes shrinkage prior ($\kappa_0 = 5.0$, blue solid curve) stabilizes within $k \le 8$ calibration trials ($\|\mathbf{R}_k - \overline{\mathbf{R}}_{\text{test}}\|_F / \|\overline{\mathbf{R}}_{\text{test}}\|_F \le 29.38\%$, reaching 3.60% asymptotically at trial 60; 36.47% at trial 8 under the unscaled source prior). This outperforms trace-matched cold-start identity initialization ($\widetilde{\mathbf{R}}_0 = \frac{\text{tr}(\mathbf{C}_1)}{C}\mathbf{I}$, orange dashed line: 47.95% error at trial 8).

As shown in Supplementary Fig. S2, initializing with trace-matched $\widetilde{\mathbf{R}}_{\text{src}}$ under $\kappa_0 = 5.0$ reduces the relative covariance error to 29.38% by trial 8 (reaching 3.60% asymptotically), enabling online covariance adaptation for zero-calibration BCIs.

S.IV. Computational Reproducibility Specifications

All experiments and benchmark evaluations were conducted in Python 3.11 with PyTorch (version 2.11.0+cu128 with CUDA 12.8 acceleration on an NVIDIA GeForce RTX 4060 Laptop GPU). Model configurations, layer dimensions, and training hyperparameters are summarized in Supplementary Table S2.

TABLE S2
Hyperparameter configurations and training specifications across all evaluation tiers.

Specification	Value / Description
EEG Sampling Rate ($f_e \to f_e^{\text{eff}}$)	100 Hz → 25 Hz (4× temporal average pooling)
fNIRS Sampling Rate (f_n)	10.33 Hz (124 samples over 12.0 s window)
Temporal Receptive Field (T_{win})	EEG: 10.0 s (1000 samples); fNIRS: 12.0 s (124 samples)
Attention Embedding Dimension (D)	64 ($d_h = 16$ across 4 heads)
Delay Prior Centers (τ_h)	$\{3.5, 4.5, 5.5, 6.5\}$ s
Delay Prior Dispersion (σ_h)	1.5 s (fixed physiological prior)
Asymmetric Post-Peak Decay Factor	$0.3\,\text{s}^{-1}$ for $\delta_{ij} > \tau_h$
Causal Mask Condition	$M_{ij} = -\infty$ for $\delta_{ij} < 0$ s ($t_{\text{nirs}} < t_{\text{eeg}}$)
Direct Stream Feature Dimension	234 (θ, α, β power + delayed HbO/HbR + slope)
Optimizer / Learning Rate	AdamW / $\eta = 3 \times 10^{-3}$, weight decay 0.05
Warmup / Scheduler	2-epoch linear warmup + cosine annealing
Batch Size / Training Epochs	Primary MA: 256 / 12 epochs; MI Benchmark & Ablation: 128 / 15 epochs
Auxiliary Loss Weights	$\lambda_{\text{deep}} = 0.2, \lambda_{\text{shallow}} = 0.2, \lambda_{\text{recon}} = 1.0, \lambda_{\text{margin}} = 0.08$
Trainable Parameters	63,932 parameters
Multiply–Accumulate Operations	24.79 Million MACs (batch size 1)
Median GPU Latency	2.22 ms (RTX 4060 Laptop GPU, P99: 4.48 ms)

S.V. Independent External Cohort Cross-Subject Validation (Buccino et al. 2016)

To evaluate the architecture on a second acquisition system and task protocol, we trained and tested NVC-DelayFormer separately under leave-one-subject-out (LOSO) cross-validation on the public Buccino et al. benchmark [5] (DOI: 10.1371/journal.pone.0146610). Its original collection reported written consent and University of Houston Institutional Review Board approval.

A. Dataset Cohort and Recording Hardware

The Buccino et al. cohort comprises $N = 15$ healthy adult participants performing sensory-motor rhythm (SMR) tasks across 3,000 trials (1,500 movement execution trials vs. 1,500 baseline rest trials):

1) **Hardware Instrumentation:** Simultaneous recordings used a wearable microEEG system (BioSignal Group; 21 scalp electrodes placed in the international 10–5 system) paired with a NIRScout 8-16 continuous-wave optical topographer (NIRx Medizintechnik; 34 optode pairs covering motor cortex at 760 nm and 850 nm wavelengths, converted to 34 HbO + 34 HbR channels via the Modified Beer–Lambert Law). This instrumentation and spatial montage are distinct from the BrainAmp and NIRScout setup of the primary TU Berlin cohort.
2) **Task Protocol and Window Duration:** Each trial spans a sustained 6.0 s block ($T_{\text{eeg}} = 600$ at 100 Hz resampled from 250 Hz; $T_{\text{nirs}} = 63$ at 10.42 Hz). This duration overlaps the specified hemodynamic delay range ($\tau_h \in [3.5, 6.5]$ s); the upper endpoint extends beyond the observed window.

B. Cross-Subject 15-Fold LOSO Evaluation

Evaluation followed 15-fold LOSO cross-validation under identical loss formulations and training hyperparameters (for each fold $s \in \{1, \ldots, 15\}$, 12 participants supplied 2,400 training trials, 2 cyclic participants supplied 400 validation trials for model selection, and 1 held-out participant supplied 200 test trials). Supplementary Table S3 details the performance across all evaluated architectures and ablation variants.

TABLE S3
CROSS-SUBJECT 15-FOLD LOSO BENCHMARK PERFORMANCE ON THE INDEPENDENT BUCCINO ET AL. (2016) COHORT ($N = 15$ PARTICIPANTS, 3,000 TRIALS).

Model Architecture	Parameters	Accuracy (%)	95% CI (%)	Macro F1 (%)
Proposed NVC-DelayFormer (Full Model)	**62,340**	**79.83 ± 13.46**	**[72.38, 87.29]**	**78.49 ± 15.76**
— Factorial [Off, Uniform] Ablation	62,340	78.03 ± 13.21	[70.72, 85.35]	76.26 ± 16.38
— Direct-Only Stream (Biophysical)	62,340	76.07 ± 13.21	[68.75, 83.38]	73.33 ± 17.61
— Latent-Only Stream (Cross-Attention)	62,340	80.20 ± 10.06	[74.63, 85.77]	79.89 ± 10.40
Authentic 2D-EEGNet [6]	2,018	74.27 ± 15.98	[65.42, 83.11]	73.89 ± 16.13
Handcrafted Shrinkage-LDA	–	85.23 ± 5.96	[81.93, 88.53]	85.13 ± 6.09
Handcrafted RBF-SVM	–	84.13 ± 6.44	[80.57, 87.70]	83.86 ± 6.81

Evaluated under 15-fold LOSO across 3,000 trials (200 trials per participant; 100 movement vs. 100 rest). Values are mean ± sample standard deviation across test participants; 95% CIs use the t_{14} distribution. Paired Wilcoxon signed-rank comparisons with the Full Model did not reach significance after Benjamini–Hochberg correction ($q \in [0.1537, 0.8469]$).

C. Findings on the Independent Cohort

Two findings emerge from this separately trained external-cohort evaluation:

1) **Cross-Hardware Generalizability:** When evaluated on the independent Buccino et al. cohort recorded with distinct instrumentation (wearable microEEG + NIRScout), NVC-DelayFormer achieves $79.83\% \pm 13.46\%$ accuracy and 78.49% Macro F1, exceeding standard 2D-EEGNet by +5.57 pp (74.27%). This indicates that the multimodal temporal fusion backbone generalizes across sensor montages and hardware specifications without architecture modification.
2) **Prior Comparison Across Hardware:** On this external sensorimotor dataset, embedding neurovascular delay priors yields higher mean classification accuracy than the unconstrained counterpart (79.83% vs. 78.03%, +1.80 pp), with the latent cross-attention stream achieving 80.20% accuracy and 79.89% Macro F1, though pairwise differences are non-significant after FDR correction (Table S3). These results indicate that the biophysical prior provides consistent numerical regularization across the evaluated recording configurations.

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